

Magma & LMFDB Mini-Course

Hunt 1: Lattices and Quadratic Forms

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We will find a non-negative integer n . Each of the four core problems below has a single non-negative integer answer n_i . The bonus is harder and does not count toward the sum. Set

$$n = n_2 - n_1 + n_4 - n_3.$$

Solutions in Magma or Sage are both welcome.

Problem 1.1. Recall the lattice Λ_1 from lecture, with Gram matrix

$$G_1 = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 1 & 6 \end{pmatrix},$$

quadratic form $Q(v) = \frac{1}{2} v G_1 v^T$ (the even convention). Let n_1 be the smallest positive integer m for which the representation number

$$r_{\Lambda_1}(m) = \#\{v \in \Lambda_1 : Q(v) = m\}$$

equals 0.

Problem 1.2. Let Λ'_1 be the second lattice in $\text{cls}(\Lambda_1)$ (the genus mate of Λ_1 , unique up to isometry since $h(\Lambda_1) = 2$). Let n_2 be the number of integers $m \in \{1, 2, \dots, 1000\}$ for which

$$r_{\Lambda_1}(m) = r_{\Lambda'_1}(m).$$

(The two classes represent the *same* integers but with potentially different multiplicities; you are counting the multiplicities at which they agree.)

Problem 1.3 (the L1 puzzle, resolved). The two even unimodular lattices of rank 16 are $E_8 \oplus E_8$ and E_{16} . Witt proved their theta series coincide:

$$\theta_{E_8 \oplus E_8}(q) = \theta_{E_{16}}(q).$$

(E. Witt, *Eine Identität zwischen Modulformen zweiten Grades*, Abh. Math. Sem. Hansischen Univ. **14** (1941), 323–337, *Satz 3*.) Yet $E_8 \oplus E_8 \not\cong E_{16}$. The *root sublattice*

$$R(\Lambda) := \langle v \in \Lambda : v \cdot v = 2 \rangle_{\mathbb{Z}}$$

sits in Λ , possibly with non-trivial index. Compute

$$n_3 = [E_{16} : R(E_{16})].$$

Problem 1.4. Let n_4 be the smallest discriminant $d \geq 2$ such that some positive-definite even ternary $\mathbb{Z}$ -lattice of rank 3 with $|\det G| = d$ has class number at least 3. (Here “even” means the Gram matrix G has even entries on the diagonal, equivalently $Q(v) = \frac{1}{2}vGv^T \in \mathbb{Z}$ for all v .)

Bonus 1 (Theta series, Hecke operators, and the Shimura lift). The difference of theta series of the two classes in the genus of Λ_1 ,

$$\theta(q) := \theta_{\Lambda_1}(q) - \theta_{\Lambda'_1}(q),$$

is a cusp form: the Eisenstein contributions cancel out.

- (a) Compute $\theta(q)$ as a q -expansion to order $O(q^{30})$.
- (b) Identify the weight, level, and character of θ . (It is of half-integral weight $3/2$; note that the LMFDB does not currently tabulate half-integral weight forms, so you will identify the space it lives in rather than look it up.)
- (c) For the genus of Λ_1 (class number 2), compute the p -neighbor adjacency matrix T_p on $\text{cls}(\Lambda_1)$ for $p = 2, 3, 5, 7$. Extract the non-trivial eigenvalue of each T_p (the trivial eigenvalue is $p + 1$, coming from the Eisenstein average). These non-trivial eigenvalues are *not* the Hecke eigenvalues of the half-integral weight form θ from (b); rather, they match the Hecke eigenvalues of its *Shimura lift* – an integer-weight modular form of weight 2. This phenomenon is a manifestation of the exceptional isomorphism between $\text{SO}^+(2, 1)$ and PSL_2 , equivalently the Eichler correspondence between ternary quadratic forms and quaternion algebras. Identify the corresponding weight-2 newform in the LMFDB and verify the eigenvalues match.